In [ ]:
```
# EXPERIMENT 3 – CREATING DATAFRAMES & ANALYZING PROPERTIES
PIN NO.: 241UDS0110
# AIM
To create Pandas DataFrames using NumPy arrays and CSV files, and analyze DataFrame properties using a new
library-visitors dataset.
```

In [ ]:
```
## Step 1: Install and Import Libraries
```

In [2]:
```
import numpy as np
import pandas as pd
```

In [ ]:
```
## Step 2: Create a 2-D Array
```

In [3]:
```
library_visitors_241UDS0110 = np.array([
    [1250, 1380, 1425],
    [970, 1040, 1125],
    [830, 905, 990]
])
library_visitors_241UDS0110
```

Out[3]:
```
array([[1250, 1380, 1425],
       [ 970, 1040, 1125],
       [ 830,  905,  990]])
```

In [4]:
```
library_visitors_241UDS0110.shape, library_visitors_241UDS0110.ndim
```

Out[4]:
```
((3, 3), 2)
```

In [ ]:
```
## Step 3: Convert the 2-D Array into a DataFrame
```

In [5]:
```
library_index_241UDS0110 = ["Central", "North", "South"]
library_columns_241UDS0110 = ["January", "February", "March"]
library_df_241UDS0110 = pd.DataFrame(
    data=library_visitors_241UDS0110,
    index=library_index_241UDS0110,
    columns=library_columns_241UDS0110
)
library_df_241UDS0110
```

Out[5]:

	January	February	March
Central	1250	1380	1425
North	970	1040	1125
South	830	905	990

In [6]:
```
library_df_241UDS0110["April"] = [1490, 1190, 1035]
library_df_241UDS0110
```

Out[6]:

	January	February	March	April
Central	1250	1380	1425	1490
North	970	1040	1125	1190
South	830	905	990	1035

In [7]:
```
new_branch_241UDS0110 = pd.DataFrame(
    {"January": [760], "February": [825], "March": [890], "April": [940]},
    index=["East"]
)
library_df_241UDS0110 = pd.concat([library_df_241UDS0110, new_branch_241UDS0110])
library_df_241UDS0110
```

Out[7]:

	January	February	March	April
Central	1250	1380	1425	1490
North	970	1040	1125	1190
South	830	905	990	1035
East	760	825	890	940

In [ ]:
```
## Step 4: Creating a DataFrame with an Existing CSV/Excel File
```

In [8]:
```
library_data_241UDS0110 = pd.read_csv("Experiment_3_241UDS0110_Library_Visitors.csv")
library_data_241UDS0110
```

Out[8]:

	PIN	Branch	Month	Visitors	Books_Issued	Study_Hours	Membership_Type
0	241UDS0110	Central	January	1250	780	420	Student
1	241UDS0110	Central	February	1380	825	465	Faculty
2	241UDS0110	Central	March	1425	860	490	Student
3	241UDS0110	North	January	970	610	350	Student
4	241UDS0110	North	February	1040	640	370	Faculty
5	241UDS0110	North	March	1125	685	390	Student
6	241UDS0110	South	January	830	520	300	Student
7	241UDS0110	South	February	905	560	325	Faculty
8	241UDS0110	South	March	990	595	345	Student

In [ ]:
```
## Step 5: Analyze the Important Properties of the DataFrame
```

In [9]:
```
library_data_241UDS0110.head()
```

Out[9]:

	PIN	Branch	Month	Visitors	Books_Issued	Study_Hours	Membership_Type
0	241UDS0110	Central	January	1250	780	420	Student
1	241UDS0110	Central	February	1380	825	465	Faculty
2	241UDS0110	Central	March	1425	860	490	Student
3	241UDS0110	North	January	970	610	350	Student
4	241UDS0110	North	February	1040	640	370	Faculty

In [10]:
```
library_data_241UDS0110.shape
```

Out[10]:
```
(9, 7)
```

In [11]:
```
library_data_241UDS0110.index
```

Out[11]:
```
RangeIndex(start=0, stop=9, step=1)
```

In [12]:
```
library_data_241UDS0110.dtypes
```

Out[12]:
```
PIN                 str
Branch              str
Month               str
Visitors          int64
Books_Issued      int64
Study_Hours       int64
Membership_Type     str
dtype: object
```

In [13]:
```
library_data_241UDS0110.columns
```

Out[13]:
```
Index(['PIN', 'Branch', 'Month', 'Visitors', 'Books_Issued', 'Study_Hours',
       'Membership_Type'],
      dtype='str')
```

In [ ]:
```
## Step 6: Information About the DataFrame
```

In [14]:
```
library_data_241UDS0110.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 9 entries, 0 to 8
Data columns (total 7 columns):
 #   Column           Non-Null Count  Dtype 
---  ------           --------------  ----- 
 0   PIN              9 non-null      str   
 1   Branch           9 non-null      str   
 2   Month            9 non-null      str   
 3   Visitors         9 non-null      int64 
 4   Books_Issued     9 non-null      int64 
 5   Study_Hours      9 non-null      int64 
 6   Membership_Type  9 non-null      str   
dtypes: int64(3), str(4)
memory usage: 636.0 bytes
```

In [ ]:
```
## Step 7: Descriptive Statistics
```

In [15]:
```
library_data_241UDS0110.describe()
```

Out[15]:

	Visitors	Books_Issued	Study_Hours
count	9.000000	9.000000	9.000000
mean	1101.666667	675.000000	383.888889
std	209.359141	120.908023	63.775866
min	830.000000	520.000000	300.000000
25%	970.000000	595.000000	345.000000
50%	1040.000000	640.000000	370.000000
75%	1250.000000	780.000000	420.000000
max	1425.000000	860.000000	490.000000

In [ ]:
```
## Step 8: Transpose Rows and Columns
```

In [16]:
```
library_data_241UDS0110.head(10).transpose()
```

Out[16]: Columns 1–8 of 10

	0	1	2	3	4	5	6
PIN	241UDS0110	241UDS0110	241UDS0110	241UDS0110	241UDS0110	241UDS0110	241UDS0110
Branch	Central	Central	Central	North	North	North	South
Month	January	February	March	January	February	March	January
Visitors	1250	1380	1425	970	1040	1125	830
Books_Issued	780	825	860	610	640	685	520
Study_Hours	420	465	490	350	370	390	300
Membership_Type	Student	Faculty	Student	Student	Faculty	Student	Student

Columns 9–10 of 10 · first column repeated

	7	8
PIN	241UDS0110	241UDS0110
Branch	South	South
Month	February	March
Visitors	905	990
Books_Issued	560	595
Study_Hours	325	345
Membership_Type	Faculty	Student

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```
## RESULT
Using NumPy and Pandas, a DataFrame was created from a 2-D array, a column and row were added, a CSV file was
loaded, and DataFrame properties, information, descriptive statistics, and transpose were examined.
```